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Proximity rapid thermal diffusion for simultaneous phosphorus and boron doping of silicon

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ABSTRACT

This study investigates simultaneous phosphorus and boron diffusion in silicon (Si) wafers using the proximity rapid thermal diffusion (PRTD). This method enables non-contact co-diffusion of dopants from spin-on dopant (SOD) sources wafers positioned on both sides of the processed wafer, forming an n^+pp^+ structure within a single rapid thermal cycle. The effects of annealing temperature and duration, pre-baking temperature, and oxygen (O_2) content in nitrogen atmosphere on dopant diffusion and sheet resistance were examined. Optimal diffusion parameters were found to be 925 °C for 180 s with pre-baking at 300 °C in the absence of O_2. Fourier transform infrared (FTIR) spectroscopy confirmed the formation of dense phosphosilicate glass and borosilicate glass layers and verified dopant incorporation into Si. Comparative device characterization demonstrated that both two-layer and three-layer PRTD configurations outperform conventional rapid thermal diffusion (RTD). The three-layer structure exhibited effective phosphorous and boron co-diffusion within a single thermal cycle, validating the feasibility of the PRTD method and its applicability for controlled multilayer doping in Si-based devices.

1. Introduction

Multilayer doped semiconductor structures represent a mature and well-established concept in electronic and optoelectronic devices based on silicon (Si). Such architectures are widely implemented in rectifying and power diodes, photodiodes, bipolar transistors and solar cell devices [1–3]. Over the past decades, continuous advancements in these fields have been driven by the demand for improved electrical performance, durability, reduced fabrication complexity, and lower manufacturing costs.

In this context, Si-based photovoltaic (PV) technologies have experienced rapid development. Advanced device concepts, such as silicon heterojunction (SHJ) and tunnel oxide passivated contact (TOPCon) solar cells, have achieved significantly enhanced power conversion efficiency (PCE) through controlled phosphorous and boron diffusion [4–7]. Similar principles of multilayer doping and junction optimization are also exploited in photodetectors, light-emitting devices, diodes and bipolar semiconductor devices. This highlights the broad technological relevance of controlled diffusion strategies across multiple application domains [8–12].

Despite differences in device functionality, many of these architectures rely on the well-established n^+pp^+ structure widely used in Si-based devices. In this configuration, the heavily doped regions (n^+ and p^+) correspond to high dopant concentrations, typically exceeding $\sim 10^{19}$ cm^{-3}, which is a standard definition in semiconductor physics and is essential for achieving low resistivity and efficient carrier transport [13,14]. Such a structure requires the formation of high-quality, well-controlled junctions to ensure optimal electrical performance. These junctions are obtained using different doping techniques, including ion implantation [15], gas-phase diffusion [16], and spin-on doping [17]. Among these, boron tribromide (BBr_3) diffusion and phosphoryl chloride ($POCl_3$) diffusion are the most widely adopted industrial methods for boron and phosphorus doping respectively. These methods provide uniform doping, stable results and proven industrial compatibility [18,19]. However, $POCl_3$ and BBr_3 diffusion processes are associated with significant safety concerns due to the use and release of highly toxic and corrosive gases. These hazards necessitate strict safety protocols and increase both operational complexity and cost. As an alternative, the solution-based spin-on dopant (SOD) technique is recognised as a promising and cost-effective approach. In the SOD process, phosphosilicate glass (PSG) or borosilicate glass (BSG) layers are formed directly on the wafer surface, followed by rapid thermal diffusion (RTD),

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enabling efficient dopant incorporation (a conventional RTD-based structure is illustrated in Fig. 1a). In recent years, high-performance TOPCon solar cells have been successfully fabricated using both boron- and phosphorus-based SOD solutions. For instance, Yang et al. fabricated TOPCon structure using commercial SOD solution supplied from Filmtronics and achieved PCE of 18.5 % while open-circuit voltage (V_{oc}) of these solar cells showed outstanding results reaching 700 mV [20]. Similarly, Park et al. achieved PCE of 17.5 % with V_{oc} of 695 mV for Si solar cells employing Filmtronics SOD solution [21]. This demonstrated performance underscores the practical relevance of SOD, which further benefits from compatibility with scalable deposition techniques such as spray coating, bar coating, screen printing, etc [22].

Despite its advantages, conventional SOD technique combined with diffusion at high temperature can introduce defects into the Si wafer, subsequently reducing performance of Si based devices. For instance, during boron diffusion, a boron-rich layer (BRL) can form on top of the p^+-doped region. The presence of this BRL is known to reduce the bulk minority-carrier lifetime by introducing defect states within the Si bulk [23]. In addition, the direct deposition of dopant solutions onto the Si surface may result in the incorporation of metallic impurities, such as Fe, Cr, Cu, and Ni [24]. These contaminants can diffuse into the Si wafer during subsequent thermal processing. Once incorporated, they form recombination centres that degrade carrier lifetime and ultimately reduce the PCE of Si-based devices.

To address this issue, Proximity Rapid Thermal Diffusion (PRTD) has been proposed as one of the effective techniques that enables successful dopant transfer [25,26]. In comparison with conventional RTD methods, PRTD is a non-contact diffusion approach in which Si wafer is placed in proximity to a dopant source wafer. Importantly, the absence of direct contact in PRTD reduces the risk of metallic impurities incorporation. Additionally, this suppresses the formation of BRL, which is commonly observed in conventional RTD processes [27]. During PRTD, heat from halogen lamps causes phosphorus pentoxide (P_2O_5) or boron oxide (B_2O_3) to evaporate from the source wafers and transport through the gas phase to the surface of the processed (undoped) Si wafer, forming a doped layer (further referred to as processed wafer) [28,29]. Currently, PRTD method is typically performed either as a one-sided diffusion process (referred to as two-layer PRTD in this study) to form n^+ or p^+ region, or as two separate diffusion steps for creation of n^+pp^+ structure [26,29,30].

In this study, a simultaneous phosphorus and boron co-diffusion is introduced to form an n^+pp^+ structure through single-step thermal annealing. This is achieved using a three-layer PRTD configuration, in which phosphorus and boron SOD sources are placed on opposite sides of the processed wafer (Fig. 1c). The series of experiments was conducted to optimize diffusion parameters, including annealing temperature and duration, pre-baking temperature, and oxygen (O_2) concentration. Subsequently, their effect on dopant incorporation and sheet resistance (R_s) was investigated. For a systematic comparison and to clearly demonstrate the advantages of the three-layer PRTD approach over conventional SOD, additional experiments were performed using two-layer PRTD (phosphorus diffusion only) as well as conventional RTD. Subsequently, simplified solar cell devices were fabricated using all three approaches. The results demonstrate that PRTD outperforms conventional RTD by maintaining superior PCE.

2. Experimental procedure

Monocrystalline Si wafers, p-type, 0.5-2 Ω cm, 330 ± 30 μm, <100> oriented were used as source and processed wafers. The wafers were cleaned in Piranha solution (mixture of sulfuric acid (H_2SO_4) and hydrogen peroxide (H_2O_2)) for 10 min and were etched in potassium hydroxide (KOH) solution for 45 min. The etching process was performed to remove saw damage. The final thickness of 170-190 μm was obtained. Then samples were dipped in hydrofluoric acid (HF, 3 %) and RCA-2 (containing hydrochloric acid (HCl), H_2O_2 and H_2O in a ratio of 1:1:6, respectively). The phosphorus solution contained orthophosphoric acid (H_3PO_4), tetraethyl orthosilicate (TEOS) and isopropyl alcohol (IPA), while boron solution contained B_2O_3, TEOS, IPA and HCl as a catalyst. Obtained solutions were stirred for 60 min and 150 min, respectively. Detailed formulation is described in Ref. [31].

Before the deposition the prepared solutions were mixed with ethanol at a 2:1 ratio to reduce the viscosity of the solutions. Subsequently, they were filtered through a PTFE 0.45 μm syringe filter to facilitate the formation of uniform and high-quality silicate films. The solutions were deposited using a spin-coating method on an SM-180-BT spin-coater at 3500 rpm rotation speed for 30 s. The source wafers were spin-coated with either phosphorus or boron SODs. For the conventional RTD samples, phosphorus and boron dopants were deposited on opposite sides of the wafer using a same technique.

All deposited layers were dried by pre-baking on a hotplate at temperatures of 100, 200, and 300 °C for 10 min to optimize the process conditions. Based on this optimization, a temperature of 300 °C was selected for all subsequent samples. All pre-baking steps were carried out in air at atmospheric pressure.

Diffusion processes were conducted through As-One RTP system (before the annealing process pyrometer was calibrated for different arrangements using thermocouple). During diffusion process the source

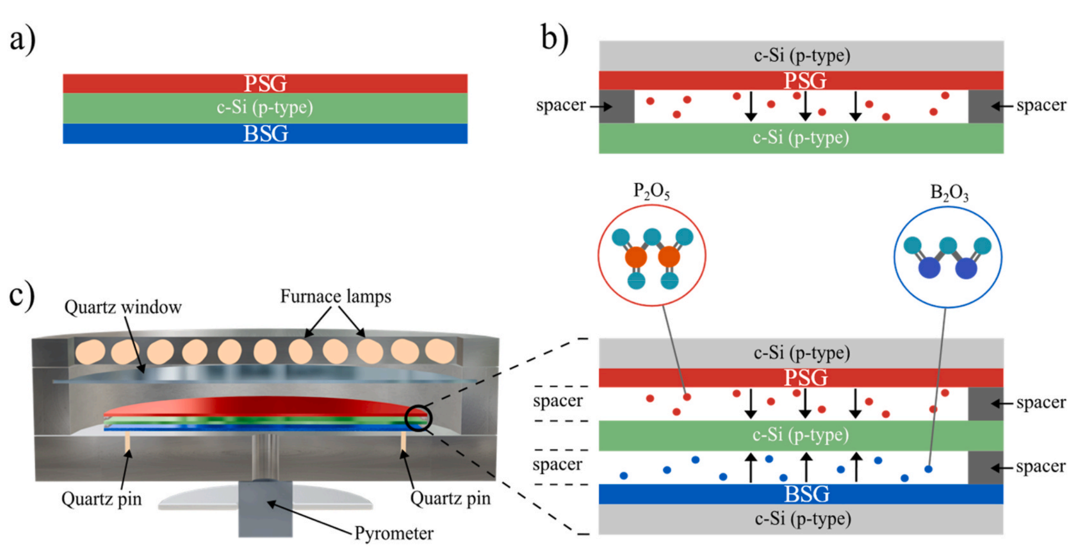

Fig. 1. (a) RTD; (b) two-layer PRTD; (c) three-layer PRTD.

and processed c-Si wafers were stacked in proximity on 0.3 mm Si spacers, thus creating a three-layer structure (Fig. 1c). Two-layer structure was obtained by stacking phosphorous doped source wafer on a clean c-Si wafer (Fig. 1b) and conventional RTD wafer was annealed separately (Fig. 1a). After diffusion process all processed and RTD wafers were dipped in HF (10 %) to remove the PSG and BSG layers. Fga

A series of preliminary experiments was conducted to select optimal parameters for three-layer PRTD. The annealing process was carried out at 825 °C, 875 °C, 925 °C and 975 °C (180 s, pre-baked at 300 °C). Subsequently, annealing duration was set to 60, 120, 180 and 240 s (925 °C, pre-baked at 300 °C). The content of O_2 in nitrogen (N_2) atmosphere was set to 0 %, 5 %, 10 % and 15 % (925 °C, 180 s, pre-baked at 300 °C). The wafers were pre-baked at 100 °C, 200 °C and 300 °C (925 °C, 180 s).

After optimization of relevant parameters, three-layer PRTD (Fig. 1c), two-layer PRTD (Fig. 1b) and conventional RTD samples (Fig. 1a) were compared to demonstrate the effect of PRTD method on performance of Si solar cell. Therefore, simplified solar cells were fabricated using processed PRTD and conventional RTD wafers. Indium Tin Oxide (ITO) layer with a thickness of 80 nm was deposited as a transparent electrode on the n-side of the processed wafers. Aluminium (Al) layer with a thickness of 1 μm was deposited as a back electrode on the p-side of processed wafers using magnetron sputtering. After the deposition, the forming gas annealing (FGA) (with 5 % H_2 and 95 % N_2 content) at 400 °C was performed to facilitate hydrogen passivation of bulk and interface defects.

The R_s measurements were conducted by a four-point probe method using Keithley Source Meter 2400 (USA) to evaluate the effect of different parameters on electrical characteristics (schematic illustration is provided in Ref. [32]). The ABET Solar System (USA) was used to measure *I-V* and *J-V* characteristics. The QE-R Enlitech (Taiwan) measuring instruments were used to study External Quantum Efficiency (EQE). Fourier transform infrared spectroscopy (FTIR) analysis was performed in the range of 400–4000 cm^{-1} to characterize the chemical composition of PSG and BSG thin films on Nicolet iS 50 spectrometer (USA).

3. Results and discussion

3.1. Optimization of diffusion parameters

Fig. 2 shows the relationship between R_s and different annealing parameters, along with the effect of pre-baking temperature. Fig. 2a shows the variation of R_s with the change of the annealing temperature from 825 °C to 975 °C. The results show that increasing the annealing temperature leads to a decrease in R_s for the n-side. Optimal R_s range is defined as a processing window that balances sufficient dopant activation with preserved junction quality [33,34]. Excessively low R_s may be associated with over-diffusion and increased recombination, whereas high R_s reflects incomplete dopant activation and reduced electrically active dopant density. At 825 °C the wide variation is present with high values of R_s, which indicates poor quality of the doped layer. As temperature increases from 825 °C to 875 °C, average R_s sharply decreases from approximately 345 Ω/□ to 85 Ω/□. A slight increase in R_s is then observed at higher temperatures, reaching about 100 Ω/□ at 925 °C and 115 Ω/□ at 975 °C. This behaviour confirms that higher annealing temperatures enhance dopant activation and incorporation by promoting increased diffusivity [35]. For the p-side, R_s initially increases slightly between 825 °C and 875 °C, followed by a gradual decrease from approximately 120 Ω/□ to 75 Ω/□ as the temperature rises to 975 °C. As it was observed temperature has a noticeable impact on the diffusion process. Although annealing at 875 °C yields near-optimal R_s values with minimal variation for the n-side, the corresponding R_s values for the p-side remain relatively high. In contrast, annealing at 925 °C provides more balanced and consistent R_s values for both n- and p-sides. Therefore, an annealing temperature of 925 °C was selected for further experiments.

The subsequent experiment was conducted with different annealing

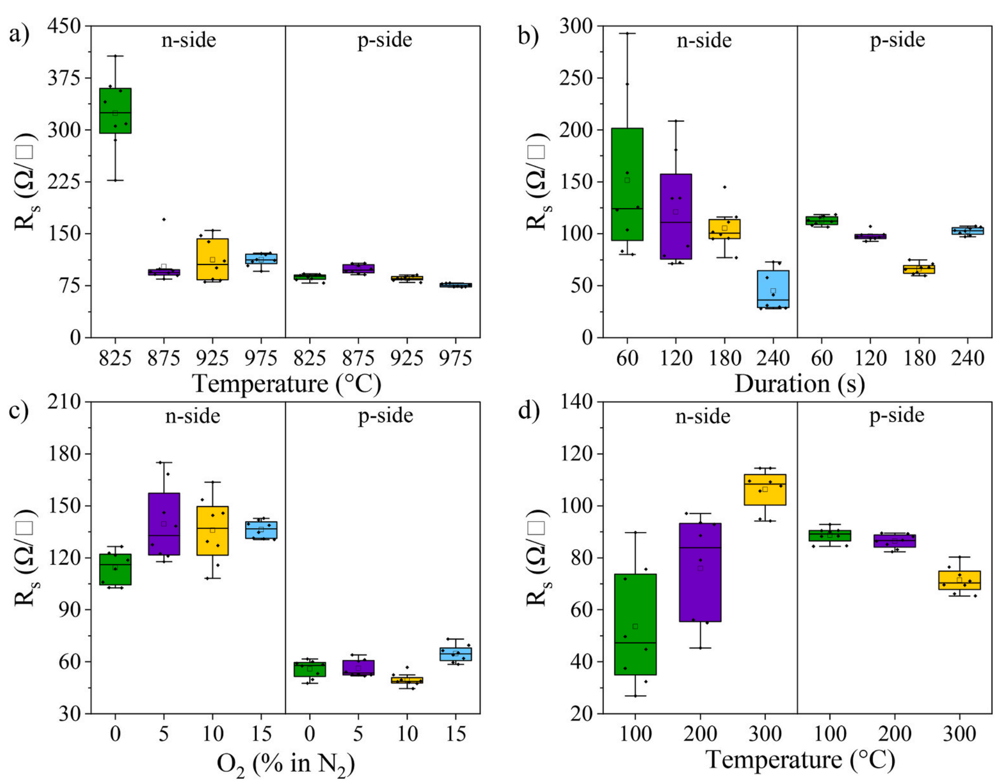

Fig. 2. Sheet resistance of the processed samples: (a) at different annealing temperatures; (b) at different annealing durations; (c) at varying concentrations of O_2 in N_2; (d) at various pre-baking temperatures.

duration to establish their effect on the resulting R_s (Fig. 2b). According to the obtained results, increase in the annealing duration from 60 to 240 s leads to a decrease in the average R_s from 125 to 35 Ω/□ for the n-side. Annealing durations of 60 and 120 s resulted in high variation and increased R_s values of 75-290 Ω/□ and 75-210 Ω/□, respectively. Increasing the annealing duration to 180 s reduces both the variation and R_s values, indicating more effective and uniform doping. A further increase to 240 s leads to excessively low R_s of 25-75 Ω/□. For the p-side, the similar reduction in R_s is observed from 60 to 180 s, followed by an increase at 240 s. The values decrease from 120 to 60 Ω/□, then rise to 100 Ω/□ at 240 s. Such an increase may be attributed to high density of phosphorus atoms evaporating from source wafer and diffusing onto the p-side, resulting in higher R_s [18]. This also explains the drop in average R_s from 100 to 35 Ω/□ on the n-side of the wafer. Annealing duration of 180 s exhibited more optimal R_s ranging from 75 to 120 Ω/□ for the n-side and 55-70 Ω/□ for the p-side. Consequently, in further experiments the duration of 180 s was selected.

Some studies propose that introducing controlled amounts of O_2 into the N_2 ambient enables the formation of thin oxide during diffusion that controls dopant incorporation and acts as a passivating layer. This helps to achieve lower and more consistent R_s and improve diffusion uniformity [29,36]. To evaluate this effect within the three-layer PRTD process, an experiment with various O_2 concentrations was conducted (Fig. 2c). For the n-side, the dependence of R_s on O_2 concentration is non-monotonic. Increasing the content of O_2 in N_2 from 0 % to 5 % results in an increase of R_s values from 100 to 130 Ω/□ to 115-175 Ω/□. Raising the O_2 concentration to 10 % results in a partial reduction to 110-165 Ω/□. However, R_s increases once more at 15 % O_2, reaching 130-145 Ω/□. A similar trend is also observed on the p-side. The R_s value increases from 45 to 60 Ω/□ to 50-65 Ω/□, subsequently decreases slightly to 45-55 Ω/□ at 10 % O_2, and increases again to 60-75 Ω/□ at 15 %. These absolute changes, however, are considerably smaller than those measured on the n-side. This behaviour possibly indicates that the p-side is substantially less sensitive to O_2 during PRTD processing. The results obtained on both sides may be explained by the rapid formation of a thin SiO_2 layer during RTD, produced even by small concentrations of O_2. This reduces the number of available incorporation sites for phosphorus and leads to non-uniform oxide thickness, thereby increasing variability and preventing any systematic improvement in R_s [37,38]. Moreover, the involvement of boron dopants in the three-layer PRTD configuration may influence the overall R_s behaviour. It has been reported that O_2 can interact with boron in crystalline Si, forming boron-oxygen-related defects that degrade electrical properties [39]. Given the negligible effect of varying O_2 concentrations and the optimal performance observed under an O_2-free environment, all subsequent experiments were carried out under 0 % O_2.

Fig. 2d demonstrates R_s values at various pre-baking temperatures for phosphorus and boron source wafers. According to the obtained results, increasing the pre-baking temperature leads to an increase of average R_s values from 45 to 110 Ω/□ for n-side. The wafers pre-baked at 100 °C exhibit a wide variation and low R_s values of 25-90 Ω/□. This probably suggests an incomplete evaporation of residual solvents which causes the presence of hydroxyl-rich, highly porous PSG layer. The presence of hydroxyl groups in silicate glasses is generally undesirable as they hinder controlled dopant diffusion [31]. Higher pre-baking temperatures promote an increase in R_s values and decrease in variability, suggesting reduced dopant concentration. This may be attributed to the formation of a more uniform SOD layer resulting from hydroxyl removal and network densification [27]. Moreover, partial evaporation of phosphorus may occur followed by its redeposition on BSG layer. However, on the p-side, the influence of the pre-baking temperature exhibits the opposite trend. As pre-baking temperature increases, the average R_s decreases from 90 to 70 Ω/□. Higher pre-baking temperatures may promote BSG densification, which enhances the incorporation of boron into Si during the diffusion. Hence, the pre-baking temperature of 300 °C was adopted for the following experiments.

By analyzing the variation of R_s, the optimal diffusion parameters were identified. The trends observed with respect to annealing temperature and duration indicated predictable dopant activation behavior. In contrast, variations in O_2 concentration in the N_2 ambience resulted in no pronounced or systematic influence on R_s for either dopant type. Notably, the dependence of R_s on pre-baking temperature exhibited non-trivial and asymmetric behavior for the n- and p-sides. Such asymmetry is a characteristic feature of the three-layer PRTD configuration, arising from the simultaneous presence of phosphorus and boron dopant sources. During the diffusion process, phosphorus- and boron-containing species may partially evaporate and redeposit on the opposite wafer side, contributing to cross-interactions between the dopant layers. Similar effects also occur under variation of other diffusion parameters, although they are less pronounced, highlighting the need for precise optimization of diffusion conditions. Overall, these observations suggest that the pre-baking temperature primarily controls dopant release and subsequent diffusion behavior through structural modifications of the SOD layers. To further investigate the origin of this behavior and to verify the proposed structural transformations induced by elevated pre-baking temperatures, a detailed analysis of the dopant layer structure was performed.

3.2. FTIR analysis of PSG and BSG films

To understand the structural origin of the observed R_s behavior, FTIR spectroscopy was employed to examine the chemical bonding and network structure of PSG and BSG layers. FTIR analysis allows identification of changes in silicate network, hydroxyl content, and bonds formation, which are known to affect dopant release and diffusion during thermal processing. Specifically, the effect of annealing and pre-baking temperatures on structural properties of PSG and BSG films of source wafers was investigated (Fig. 3).

Fig. 3a shows the IR spectra obtained for PSG films after pre-baking at 100-300 °C. They reveal OH group stretching modes (3400 cm^{-1}), Si-O-Si asymmetric stretching (1080 cm^{-1}), Si-O bending (820 cm^{-1}), Si-O-Si rocking (450 cm^{-1}), Si-OH vibrations (950 cm^{-1}) and P-O stretching vibration (1200 cm^{-1}) bands [31]. Increase in pre-baking temperature leads to a weakening of Si-OH and OH bands, suggesting a reduction in hydrogen bonding. The increase in the intensity and a shift to higher wavenumbers of Si-O-Si stretching peak and Si-O bending peaks by increasing the temperature indicates network polymerization or densification. The pre-baking temperature influences the weakening of P-O shoulder, which may reflect enhanced network condensation of pre-baked samples and a partial removal of phosphorus-containing species in the network structure. Similar behaviour has been reported previously for pre-baked PSG films [31]. These observations are consistent with the trends in R_s results shown in Fig. 2d.

Fig. 3b demonstrates IR spectra for BSG films pre-baked at 100-300 °C. The IR spectra reveal peaks at ~1075 cm^{-1}, 945 cm^{-1}, ~460 cm^{-1}, and ~1400 cm^{-1} [27,40]. The absorption band at ~1075 cm^{-1} can be assigned to tri-, tetra-, and pentaborate groups (BO_3 and BO_4) along with Si-O-Si asymmetric stretching bands. The band at ~830 cm^{-1} corresponds to the O-Si-O symmetric stretching vibration. The peak at approximately 1400 cm^{-1} is ascribed to the B-O symmetric stretching of trigonal BO_3 units [41]. The presence of a Si-OH bands at ~945 cm^{-1} indicates incomplete condensation. A broad band at approximately ~3400 cm^{-1} suggests the presence of OH group stretching modes, signifying that non-bridging oxygens are linked to hydrogen. When the pre-baking temperature is increased, a weakening of Si-OH and OH bands is observed. This is attributed to the removal of organic residuals that were not completely evaporated at lower temperatures. Increase in pre-baking temperature leads to the increase in the intensity of Si-O-Si stretching peak and Si-O bending peak (820 cm^{-1}), suggesting the densification and increase of the layer thickness. The shift of the Si-O-Si band towards lower wavenumbers with increasing the temperature may be associated with enhanced boron incorporation into the silicate

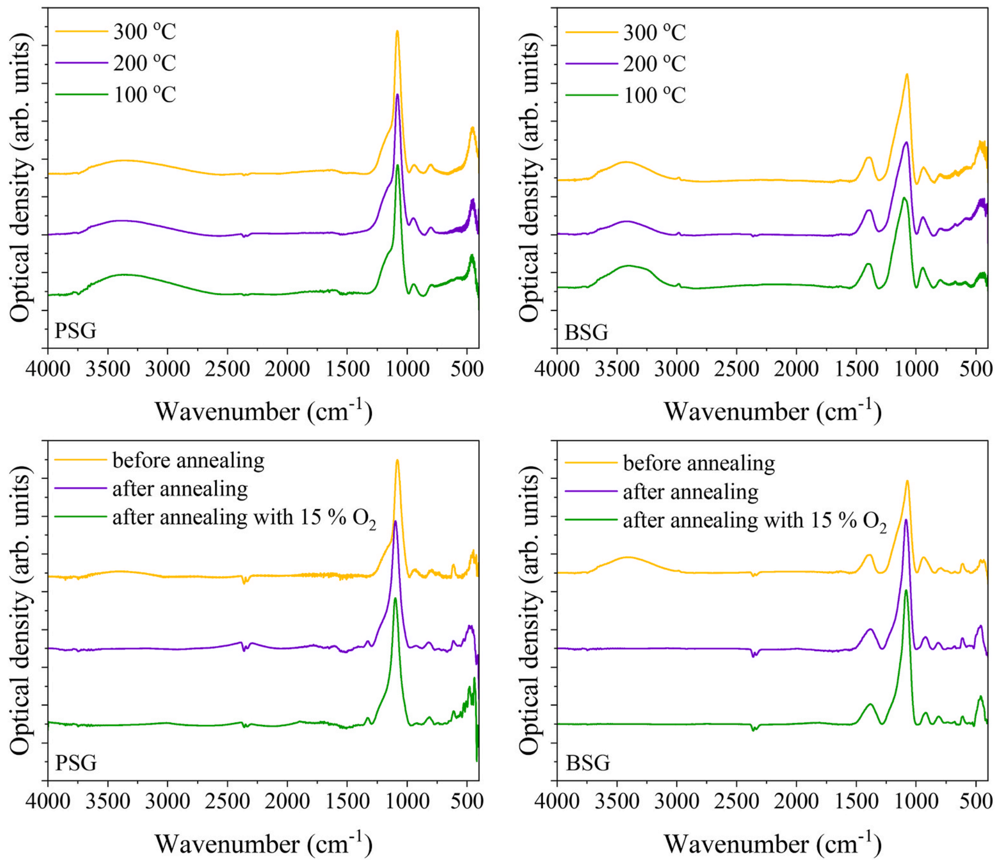

Fig. 3. FTIR spectra of source wafers: (a) PSG films after pre-baking at 100-300 °C; (b) BSG films after pre-baking at 100-300 °C; (c) PSG films before, after annealing (925 °C, 180 s, 0% O_2) and after annealing with 15 % O_2; (d) BSG films before, after annealing (925 °C, 180 s, 0% O_2) and after annealing with 15 % O_2.

network. As pre-baking temperature increases, the B-O band shifts to lower wavenumbers from ~1400 cm^{-1} to ~1390 cm^{-1}. This characteristic change may be due to the densification of BSG layer. However, BSG films show subtle changes in the intensity of B-O region, indicating better integration of boron into the Si-O network [27]. These observations support previously made assumptions on the effect of pre-baking temperatures on R_s.

Fig. 3c depicts IR spectra, comparing PSG films before annealing, after annealing and after annealing with 15 % O_2 introduced into the N_2 ambient, both performed at 925 °C for 180 s. After annealing, OH bands at ~3400 cm^{-1} and Si-OH bands at ~945 cm^{-1} disappear, indicating dehydroxylation reactions (removal of hydroxyl groups), which progressively transform the dried gels into PSG. Moreover, new P=O bands appear at 1330 cm^{-1}. Annealing leads to the shifts in the P-O (1200 cm^{-1}) and Si-O (845 cm^{-1}) bands to higher wavenumbers. This suggests interactions between Si-O and P=O bands (1330 cm^{-1}) [40,42], resulting in Si-O-P bond formation, indicating the phosphorus diffusion into Si [43]. Consequently, the annealed films may exhibit higher density compared to the as-deposited films. The shoulder associated with P-O bands almost disappears, indicating more complete densification and possible reduction in dopant content of the samples after annealing [44]. After annealing, the Si-O-Si rocking peak at ~454 cm^{-1} intensifies, which may indicate the removal of hydroxyl groups and the formation of a more ordered and densified SiO_2 layer. This suggests a higher concentration of Si-O-Si bonds and enhanced structural stability of the layer. Annealing in the presence of 15 % O_2 results in a further, though relatively small, increase in the Si-O-Si rocking band. This suggests additional O_2 incorporation into the Si network and promoting SiO_2 formation. However, the introduction of O_2 during annealing does not lead to significant changes compared to annealing without O_2.

Fig. 3d demonstrates IR spectra, comparing BSG films before annealing, after annealing without O_2 and after annealing with 15 % O_2. For BSG films annealing promotes structural relaxation, with boron atoms potentially acting as sites for oxygen trapping or clustering within the BSG network. The bands around 800 cm^{-1}, associated with O-Si-O bands, also exhibit increased intensity after annealing. The OH peak at ~3400 cm^{-1} and Si-OH peak at 934 cm^{-1} disappear after annealing, indicating the loss of hydroxyl groups. Additionally, the band at 917 cm^{-1}, corresponding to the B-O-Si stretching mode of BO_4 tetrahedral units, emerges, suggesting the stabilization of BSG network. The Si-O-Si asymmetric stretching mode at ~1075 cm^{-1} shifts to higher wavenumbers, signifying an increase in SiO_2 thickness and densification of the SOD layer. The B-O stretching mode, located at 1387 cm^{-1} decreases in intensity and shifts to lower wavenumbers, indicating layer densification [27]. The intensity of the Si-O-Si bending vibration at 460 cm^{-1} increases after annealing, suggesting greater oxygen incorporation into the Si lattice and formation of SiO_2 layer [44]. The introduction of O_2 into the N_2 ambient during annealing leads to a further increase in this peak, which can be attributed to a slight rise in oxygen concentration, as observed in PSG films. However, the addition of O_2 does not result in additional structural changes, as the spectra remain largely similar to those obtained after annealing in N_2 alone. This observations are consistent with the R_s measurements, confirming the insignificant effect of O_2 addition during annealing.

FTIR analysis revealed that the pre-baking temperature affects the reduction of hydrogen bonding in both PSG and BSG layers. Elevated pre-baking temperatures promote network condensation in pre-baked samples for n-side and partially remove phosphorus-containing species from the network structure along with hydroxyl groups. In BSG layers, densification and improved boron incorporation were observed. Furthermore, FTIR analysis of the annealing effect demonstrates the further elimination of residual hydroxyl groups, facilitating the conversion of the SOD gel into PSG and BSG layers. Additionally, the formation of Si-O-P and B-O-Si bonds confirm phosphorus and boron diffusion into Si network and indicates stabilization of silicate glass. These structural modifications correlate with the trends observed in R_s measurements, indicating a direct relationship between film structure and dopant diffusion behaviour.

3.3. Device performance comparison (PRTD and RTD)

Prior to the solar cells fabrication, the R_s (Fig. S1) and dark *I-V* (Fig. S2) characteristics of PRTD processed wafers and RTD wafers were evaluated. Both three-layer and two-layer PRTD configurations exhibit lower and more uniform R_s on the n-side compared to RTD, indicating improved dopant incorporation and uniform junction formation. On the p-side, the R_s obtained for the three-layer PRTD is slightly higher than that of conventional RTD and remains closer to the optimal range for device operation.

Dark *I–V* characteristics (Fig. S2) were further analysed to assess junction quality. All samples exhibit clear rectifying behaviour, indicating proper diode formation. Both two-layer and three-layer PRTD samples exhibit lower reverse-bias dark currents and reduced leakage compared to RTD samples, indicating improved junction quality and reduced recombination pathways. The three-layer PRTD demonstrates the lowest dark current in the low-bias region, confirming that the simultaneous co-diffusion of phosphorus and boron does not degrade junction quality; instead, it leads to a comparably improved diode behaviour. Therefore, dark *I-V* results validate the structural and dopant activation improvements achieved through PRTD.

To verify the beneficial effect of PRTD, simplified solar cells were fabricated using three-layer PRTD, two-layer PRTD, and RTD methods (structure is shown in Fig. 4a).

The *J–V* characteristics of the most efficient cells under illumination (1000 W/m^2) from each fabrication method are presented in Fig. 4a. Both two-layer and three-layer PRTD samples exhibit higher V_{oc} and short-circuit current density (J_{sc}) values compared to RTD. This improvement indicates reduced recombination losses in PRTD cells. Both V_{oc} and J_{sc} values for two-layer PRTD are higher than three-layer PRTD. This behaviour can be attributed to differences in dopant incorporation mechanisms between the two configurations. In the two-layer PRTD, phosphorus diffusion leads to the formation of an explicit n$^+$ region, while the p-type base remains largely unaffected. In contrast, in the three-layer PRTD configuration, simultaneous diffusion of phosphorus and boron from opposite sides may result in possible outdiffusion of boron and partial cross-contamination during annealing. This can lead to less optimal p$^+$ region formation and increased recombination, thereby slightly reducing V_{oc} and J_{sc}.

Prior to the discussion of statistical device performance, it is important to emphasize that the primary objective of this study is to assess the feasibility of the proposed diffusion method. The resulting PCE values are therefore lower than those of state-of-the-art Si solar cells due to the absence of advanced device features. The statistical performance metrics for solar cells comparing three-layer PRTD, two-layer PRTD and RTD are illustrated in Fig. S3. The results obtained by RTD shows the subsequent cell performance parameters: V_{oc} is 514 mV, J_{sc} is 21.7 mA/cm^2, and fill factor (FF) is 82 %. Three-layer PRTD demonstrates higher values: V_{oc} is 534.5 mV, J_{sc} is 22.9 mA/cm^2, and FF is 77.75 %. A two-layer PRTD displays PV parameters with V_{oc} of 538 mV, J_{sc} of 24.2 mA/cm^2, and FF of 82 %. These observations confirm the improved carrier collection and junction quality achieved through the PRTD process. The two-layer PRTD exhibits the highest overall performance with maximum PCE of 11 %. However, the three-layer configuration achieves a notable maximum PCE of 10.7 % relative to RTD with maximum PCE of 9.9 %, demonstrating the reduction of recombination sites. These results validate the effectiveness of the PRTD approach for simultaneous formation of n$^+$ and p$^+$ regions in Si solar cells. The comparative analysis of PV parameters (Fig. S3) clearly indicates that both PRTD configurations outperform conventional RTD.

It is evident that both PRTD structures outperform RTD, which may be attributed to the potential presence of BRL and metallic impurities from direct deposition of boron SOD [23,24]. To further investigate this effect, EQE measurements were conducted for all structures (Fig. 4b).

FGA was applied as a post-treatment step to passivate defects and reduce recombination via hydrogenation of interface and bulk states. However, its effect is limited without additional surface passivation layers such as SiN_x. Therefore, all EQE curves are lower than those typically observed for standard solar cells. In the short-wavelength range (300-500 nm), all samples show a rapid increase in EQE with a similar pattern, reflecting effective light absorption at the front side of the solar cell. After reaching the maximum values, all curves gradually decrease due to reduced absorption coefficient of the Si. In the long-wavelength range (600-1000 nm), the EQE values follow a decreasing trend from two-layer PRTD and three-layer PRTD samples to the RTD sample, with the two-layer PRTD showing the highest response. This behaviour can be attributed to increased recombination at the rear-side of the cell, which is more pronounced for RTD samples. Therefore, it confirms the possible incorporation of metallic impurities from the boron dopants and BRL formation. The observed behaviour is consistent with the *J-V* measurements, further confirming better performance of PRTD samples compared to the conventional RTD.

The measured electrical characteristics (dark *I-V*, illuminated *J-V* and R_s) demonstrate that the PRTD process significantly improves carrier collection and overall PV performance relative to conventional RTD.

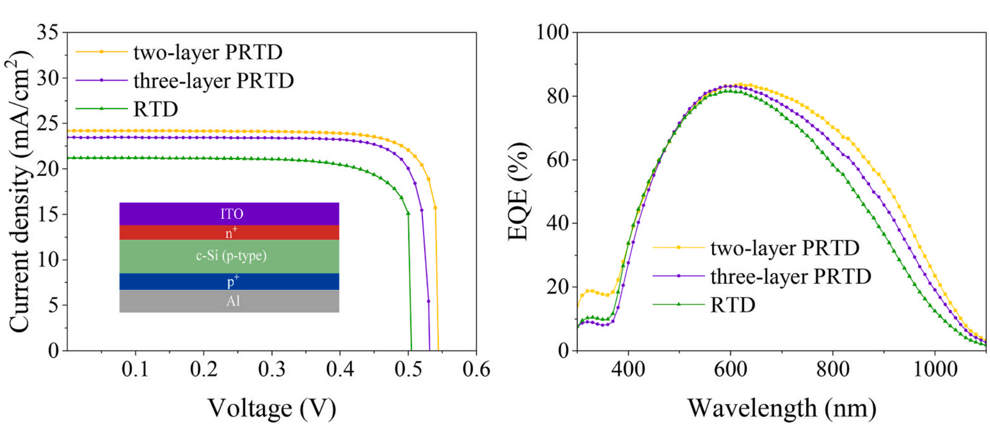

Fig. 4. a) Illuminated current density-voltage (*J-V*) characteristics of processed and conventional Si samples based on different diffusion methods and a schematic representation of Si solar cell after the deposition of ITO and Al layers on the obtained n$^+$pp$^+$ structure; b) EQE measurements for three-layer PRTD, two-layer PRTD processed and conventional RTD.

The two-layer PRTD consistently achieves the highest statistical performance metrics including PCE, which can be attributed to reduced recombination and the absence of boron dopant. EQE measurements further support these findings, demonstrating superior carrier collection in PRTD samples. The slightly lower performance of the three-layer configuration compared to the two-layer is presumably associated with BRL formation and potential metallic impurities (such as Fe, Cr, Cu, and Ni [24]) on the p-side, although it still outperforms RTD in all key metrics. Overall, this study confirms that the PRTD approach enables more controlled and uniform simultaneous formation of n^+ and p^+ regions, thereby improving device performance and demonstrating its superiority over conventional RTD methods.

The solar cells fabricated in this study employ a simplified device architecture intended to evaluate the effectiveness of the PRTD approach. While advanced features of modern high-efficiency solar cells - such as surface texturing, advanced anti-reflection coatings, and passivation layers - were not incorporated at this stage, the obtained PV parameters reliably reflect the impact of the diffusion process itself. The primary objective of this study was to demonstrate the feasibility of PRTD for forming an n^+pp^+ structure, rather than to maximize PCE. Future work should focus on integrating PRTD method into the fabrication of electronic and optoelectronic devices with advanced passivation layers, such as TOPCon bifacial solar cells, diodes and other multilayer semiconductor devices.

4. Conclusion

This work demonstrates the feasibility of PRTD for the simultaneous incorporation of phosphorus and boron into Si wafers, enabling the formation of an n^+pp^+ structure within a single rapid thermal processing cycle. A systematic investigation was conducted to examine the effects of annealing temperature and duration, pre-baking temperature, and O_2 concentration on dopant incorporation and R_s. Optimal diffusion conditions were identified at 925 °C for 180 s with a pre-baking temperature of 300 °C under O_2-free ambient conditions.

FTIR analysis revealed that pre-baking and annealing promote hydroxyl removal, network polymerization, and densification of PSG and BSG films, accompanied by the formation of Si–O–P and B–O–Si bonds. These structural modifications provide a consistent explanation for the observed R_s behaviour, highlighting the strong correlation between SOD film chemistry and dopant diffusion during PRTD.

Comparative device characterization indicated that both two-layer and three-layer PRTD configurations outperform conventional RTD in terms of PV performance. Measured electrical characteristics further confirmed improved junction quality and carrier collection in PRTD-based devices. These observations were further confirmed by EQE measurements. Therefore, three-layer configuration enabled uniform and effective co-diffusion of phosphorus and boron, resulting in superior device performance and confirming the viability of the PRTD approach.

Overall, three-layer PRTD represents a technologically feasible alternative to conventional RTD for phosphorus and boron co-diffusion in Si. The demonstrated process compatibility and structural control suggest strong potential for integrating this method into advanced Si device fabrication.

CRediT authorship contribution statement

Aiganym Suleimenova: Writing – original draft, Investigation, Formal analysis, Data curation. **Dina Akhmetova:** Writing – original draft, Visualization, Investigation, Data curation. **Sofiya Dvornikova:** Writing – original draft, Validation, Investigation, Data curation. **Aizhan Kusainova:** Writing – review & editing, Methodology. **Kair Nussupov:** Supervision, Funding acquisition. **Assanali Sultanov:** Writing – review & editing, Resources, Conceptualization. **Nurzhan Beisenkhanov:** Writing – review & editing, Supervision, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.mssp.2026.110778.

Data availability

Data will be made available on request.

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